

9	(a)	A nucleon is a proton or neutron. [B1] A nucleus is the center of an atom consisting of protons and neutrons. [B1] Nuclides are nuclei that have the same number of protons and neutrons. (or same proton and nucleon number.) [B1]	
	(b)		
	(i)	It means a carbon nucleus/atom with <u>6 protons</u> and <u>7 neutrons/13 nucleons</u> [B1]	
	(ii)	Energy of γ = equivalent mass-energy of reactants – equivalent mass-energy of $^{13}_6\text{C}$ Energy of γ = $(9.0150 + 4.0040 - 13.0075)\text{u} \times c^2$ [M1] = $1.71 \times 10^{-12} \text{ J}$ Equation 1 is not valid as energy of γ is lower than $8.8 \times 10^{-12} \text{ J}$ [A1] Energy of ^1_0n = $(9.0150 + 4.0040 - 12)\text{u} \times c^2$ [M1] = $1.52 \times 10^{-10} \text{ J}$ Equation 2 is valid as ^1_0n has energy higher than $8.8 \times 10^{-12} \text{ J}$. [A1]	
	(c)		
	(i)	Decay constant λ is the probability of decay per unit time of a nucleus [B1] $t_{1/2} = \frac{\ln 2}{\lambda}$ [B1]	
	(ii)	$A = A_0 e^{-\frac{\ln 2}{t_{1/2}} t}$ $44490 = 76300 e^{-\frac{\ln 2}{t_{1/2}}(1)}$ → $t_{1/2} = 1.29 \text{ days}$ $25940 = 44490 e^{-\frac{\ln 2}{t_{1/2}}(1)}$ → $t_{1/2} = 1.28 \text{ days}$ [M2 for each data point, if choose low activity such as day 14 and 15, will not get 1.29 days – penalize for poor choice of data] Average = $(1.28 + 1.29)/2 = 1.29 \text{ days}$ [A1 awarded only if average taken]	
	(iii)	$A = A_0 e^{-\frac{\ln 2}{t_{1/2}} t}$ $\ln A = -\frac{\ln 2}{t_{1/2}} t + \ln A_0$ [M1] Plot $\ln A$ against t [A1] The gradient is $-\frac{\ln 2}{t_{1/2}}$ and can be used to calculate the half-life. [A1]	
	(iv)	As the half-life is 1.29 days, <u>a few hours difference will result in a significant change in the activity.</u> [B1]	
	(v)	Radioactive decay is random. [B1] At such low activity, the actual activity might deviate from the expected activity by a large percentage, resulting in errors in calculation. [M1] Hence the statement is not accurate. [A1]	

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